

# Amandi Athukorala

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## Summary

Software Engineering undergraduate with hands-on experience in full-stack development, machine learning systems, cloud technologies, and DevOps practices. Skilled in designing and developing RESTful APIs, scalable backend services, data-driven applications, and cloud-native solutions using modern software engineering tools and frameworks. Experienced with containerization, CI/CD automation, Kubernetes, and machine learning deployment workflows, with a strong interest in building reliable and impactful software systems across the complete development lifecycle.

## Education

**University of Sri Jayewardenepura**  
*Bachelor of Computing Honours in Software Engineering*

Colombo, Sri Lanka  
May 2023 – Present

## Achievements and Publications

**Research Publication – JCOS 2026 (Published & Presented)** May 2026  
*1st Jayewardenepura Computing Symposium (JCOS)* University of Sri Jayewardenepura, Sri Lanka

- Published and presented a peer-reviewed research paper titled "Explainable Multi-Objective and Energy-Aware Sequence-Based Automated Refactoring in Open-Source Java: A Systematic Review" at JCOS 2026.

**WSO2 Hacktoberfest 2025 – Open Source Contributor** 2025  
*Hacktoberfest Open Source Program* WSO2

- Contributed to open-source projects under real-world development workflows, improving understanding of APIs and collaborative software engineering.

## Projects

**Leftly: Surplus Food Ordering Platform** | *Next.js, TypeScript, Express.js, PostgreSQL*

- Developed a full-stack surplus food marketplace to reduce food waste by connecting sellers with customers to purchase near-to-expiry food products at discounted prices.
- Built RESTful APIs using Express.js, Prisma ORM, and PostgreSQL to manage users, products, orders, reviews, complaints, and seller operations.
- Implemented real-time chat and order updates using Socket.io to improve communication between customers and sellers.
- Developed secure application features using JWT authentication, role-based access control, and Zustand for state management.

**TrustSpot: Trustworthy Place Discovery Platform** | *Java, Spring Boot, React, TypeScript, H2 Database*

- Developed a microservices-based place discovery platform using Spring Boot to manage authentication, place listings, and review services.
- Built RESTful APIs using Spring Data JPA for managing users, places, reviews, ratings, and business logic.
- Implemented JWT authentication and integrated React frontend components with backend microservices.
- Containerized services using Docker Compose and developed automated tests using JUnit and Mockito.

**Fraud Detection ML on Kubernetes** | *Python, FastAPI, Scikit-learn, Docker, Kubernetes, AWS S3*

- Developed an end-to-end fraud detection system using machine learning pipelines to classify fraudulent credit card transactions with automated preprocessing and model training.
- Built a FastAPI-based prediction service to provide real-time fraud predictions by loading trained models from AWS S3 storage.
- Implemented Scikit-learn pipelines with feature preprocessing and Random Forest classification, evaluating model performance using classification metrics.
- Containerized and deployed the application using Docker and Kubernetes with scalable Deployment and Service configurations.

**Dataset Drift Monitoring System** | *Python, Scikit-learn, DVC, MLflow, GitHub Actions*

- Developed an end-to-end MLOps pipeline for churn prediction with automated dataset drift detection and reproducible model training workflows.

- Implemented dataset versioning and pipeline orchestration using DVC to manage preprocessing, drift analysis, and model retraining stages.
- Built statistical drift detection using KS Test, mean difference, and variance analysis to identify distribution changes across datasets.
- Integrated MLflow and DagsHub for experiment tracking and automated CI/CD workflows using GitHub Actions.

**PromptLab: LLM Prompt Testing & Evaluation Platform** | *Python, Streamlit, Ollama, Prompt Engineering*

- Developed an interactive platform for comparing prompt engineering strategies and evaluating generated responses using local language models.
- Implemented multiple prompting techniques including zero-shot, direct instruction, and step-by-step prompting workflows.
- Built an LLM-as-a-Judge evaluation pipeline to assess response quality using both heuristic metrics and model-based scoring.
- Integrated local LLM inference using Ollama and implemented experiment logging and visualization using Streamlit dashboards.

## Leadership and Volunteer Experience

**Association of Computing Students – University of Sri Jayewardenepura**  
*Assistant Secretary*

Apr 2024 – Jun 2025

- Supported administrative coordination and event planning for computing initiatives, including workshops and student activities.

**IEEE Student Branch – University of Sri Jayewardenepura**  
*Secretary – IEEEExtreme 17.0 Organizing Committee*

Sep 2023 – Nov 2023

- Coordinated documentation, communication, and operational tasks for a 24-hour programming competition involving student participants.

**IEEE Computer Society Student Branch Chapter**  
*Design Marketing Team Member*

Apr 2025 – Oct 2025

- Created promotional materials and supported technical event initiatives to increase awareness among student communities.

## Technical Skills

**Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML/CSS, Shell Scripting  
**Frontend Development:** React, Next.js, React Native, Tailwind CSS, Zustand  
**Backend Development:** Node.js, Express.js, Spring Boot, Spring Data JPA, FastAPI, RESTful APIs, WebSockets  
**Databases & ORM:** PostgreSQL, MySQL, MongoDB, H2 Database, Prisma ORM  
**Machine Learning & AI:** Scikit-learn, TensorFlow, CNN, Pandas, NumPy  
**MLOps & Data Engineering:** MLflow, DVC, Feature Engineering, Dataset Drift Detection, Model Deployment  
**Cloud & DevOps:** AWS (EC2, S3), Docker, Docker Compose, Kubernetes, GitHub Actions, CI/CD  
**Testing & Software Engineering:** JUnit, Git, Microservices, Agile Development, OOP, Design Principles